

CONFORMAL PREDICTION

IA Frameworks

Joseba Dalman

UNCERTAINTY QUANTIFICATION IN DEEP LEARNING

- Can we trust the predictions of our models?
- Under what circumstances?
- To what extent?

CLASSICAL UQ TECHNIQUES

- Bayesian methods
- Ensembling
- Dropout
- ...



NON POST-HOC

WITHOUT GUARANTEES

CONFORMAL PREDICTION:

OBJECTIVE AND GUARANTEE

Given: a predictor $\hat{f}: \mathcal{X} \rightarrow \mathcal{Y}$
and a nominal error rate α .

Build: $\hat{C}_\alpha: \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$

With the following guarantee:

$$\mathbb{P}(Y_{\text{test}} \in \hat{C}_\alpha(X_{\text{test}})) \geq 1 - \alpha$$

OUTLINE

→ Conformal Regression

- Theorem
- Jackknife+ / CV+
- CQR

→ Conformal Classification

- LAC
- APS / RAPS

CONFORMAL REGRESSION

Given: $\hat{f}: \mathcal{X} \rightarrow \mathcal{Y}$ and α .

Build: $\hat{C}_\alpha: \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$

With the following guarantee:

$$\mathbb{P}(Y_{\text{test}} \in \hat{C}_\alpha(X_{\text{test}})) \geq 1 - \alpha$$

CONFORMAL REGRESSION

Given: $\hat{f}: \mathcal{X} \rightarrow \mathbb{R}$ and α .

Build: $\hat{C}_\alpha: \mathcal{X} \rightarrow \mathcal{P}(\mathbb{R})$
usually an interval!

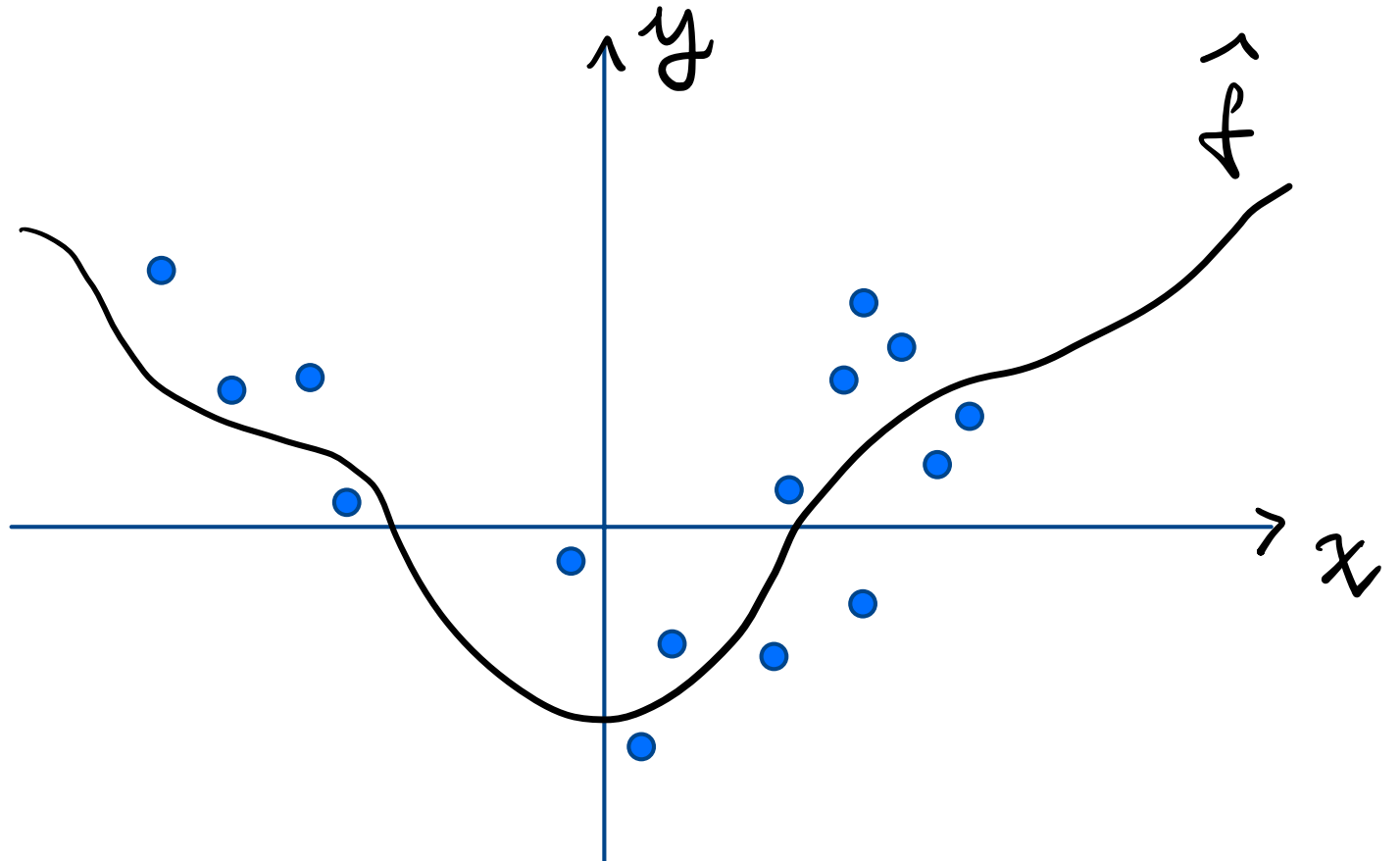
With the following guarantee:

$$\mathbb{P}(Y_{\text{test}} \in \hat{C}_\alpha(X_{\text{test}})) \geq 1 - \alpha$$

CALIBRATION

We use a calibration set

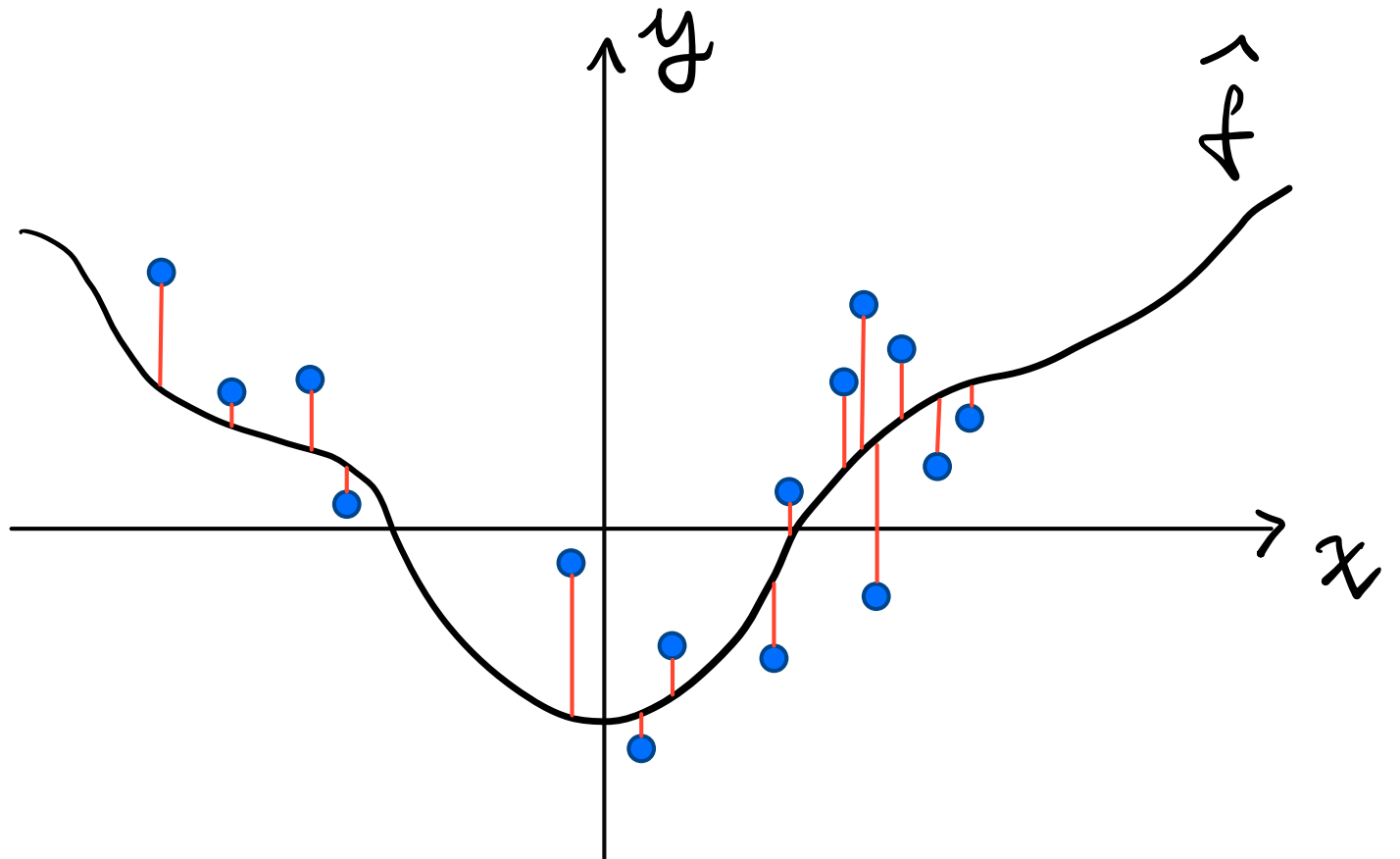
$$D_{\text{calib}} = \{(x_i, y_i)\}_{i=1}^n$$



CALIBRATION

We measure the **scores** (errors)

$$S_i = |y_i - \hat{f}(x_i)|$$



CALIBRATION

We compute:

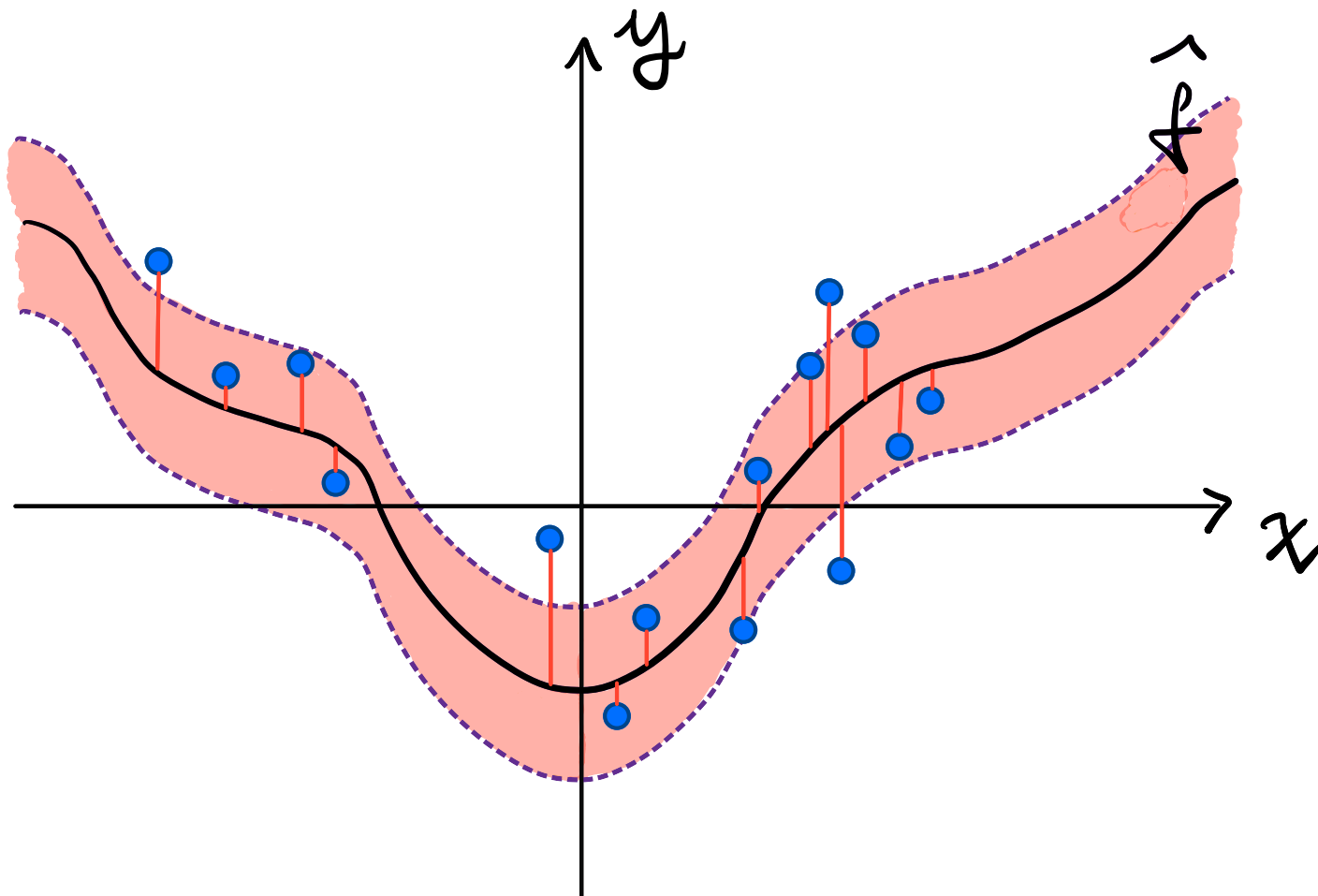
$S_\alpha :=$ the $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ -th

quantile of the scores $S_1, \dots, S_n$

i.e. the $\lceil (n+1)(1-\alpha) \rceil$ -th
smallest score.

CALIBRATION

We predict $\hat{C}_\alpha(x) = [\hat{f}(x) - \delta_\alpha, \hat{f}(x) + \delta_\alpha]$



EXERCISE

1. Write down the pseudocode of the Split Conformal Regression algorithm
2. Should we add any constraints on the nominal error rate α ?

Hint: there may be a problem when choosing the " $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score"

GUARANTEE

Theorem .- Given a calibration set

$\{(x_i, y_i)\}_{i=1}^n$ and a test point (x_{n+1}, y_{n+1}) ,

if the scores $S_i = |y_i - \hat{f}(x_i)|$ are exchangeable, then the interval

$$\hat{C}_\alpha(x_{n+1}) = [\hat{f}(x_{n+1}) - \delta_\alpha, \hat{f}(x_{n+1}) + \delta_\alpha]$$

satisfies:

$$\mathbb{P}(y_{n+1} \in \hat{C}_\alpha(x_{n+1})) \geq 1 - \alpha$$

EXCHANGEABILITY

The random variables $(X_1, X_2, \dots, X_n)$ are exchangeable if:

For any $\pi \in \left\{ \begin{array}{c} \text{permutations of} \\ \{1, \dots, n\} \end{array} \right\}$

$$X_{\pi(1)}, \dots, X_{\pi(n)} \stackrel{\text{law}}{=} X_1, \dots, X_n$$

Exercise. - Give an example of an array of r.v.s that are exchangeable but not i.i.d.

EXCHANGEABILITY

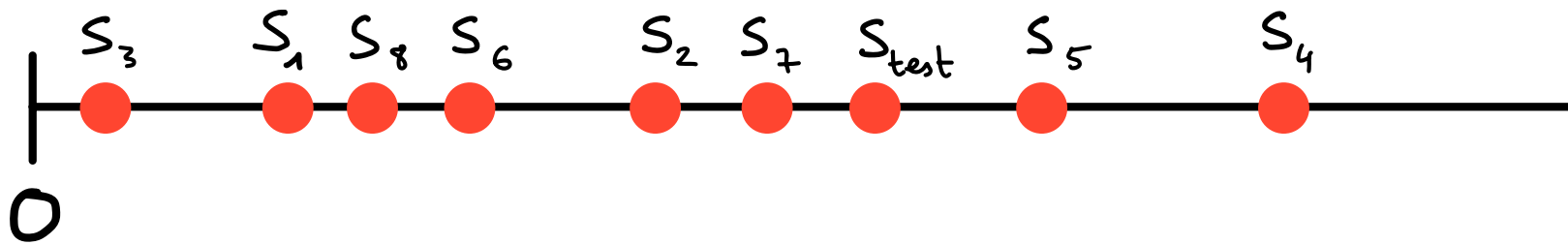
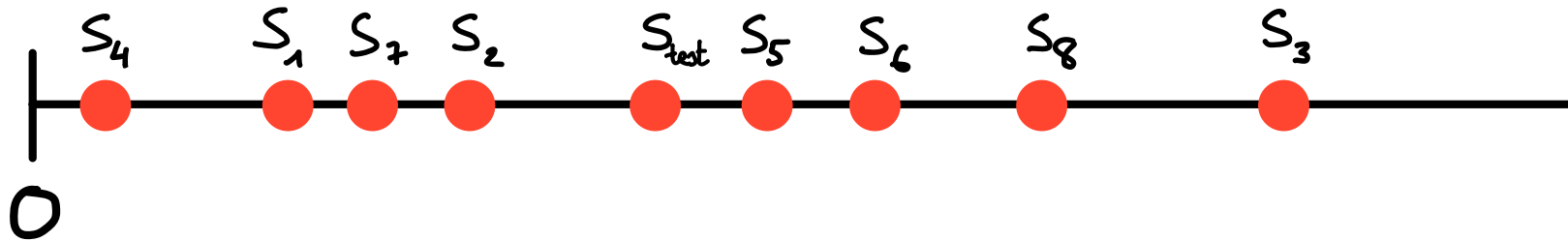
The random variables $(X_1, X_2, \dots, X_n)$ are exchangeable if:

For any $\pi \in \left\{ \begin{array}{c} \text{permutations of} \\ \{1, \dots, n\} \end{array} \right\}$

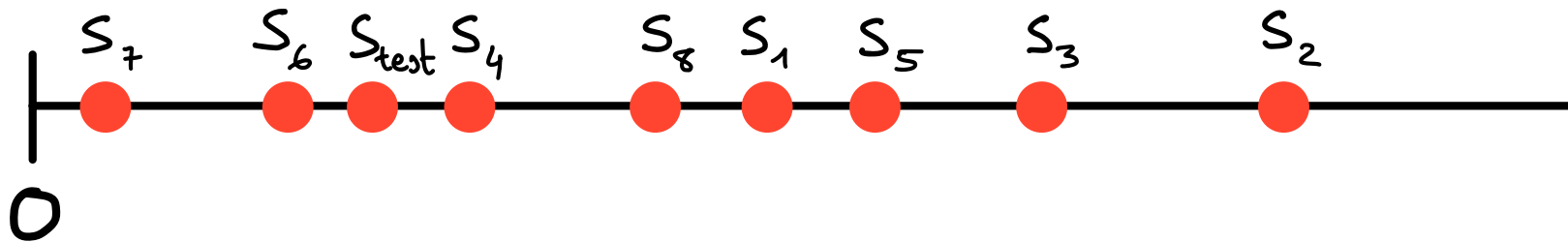
$$X_{\pi(1)}, \dots, X_{\pi(n)} \stackrel{\text{law}}{=} X_1, \dots, X_n$$

Example: $Z_1, \dots, Z_n$ i.i.d. $\Rightarrow X_i = \frac{Z_i}{\sum_{j=1}^n |Z_j|}$
exchangeable

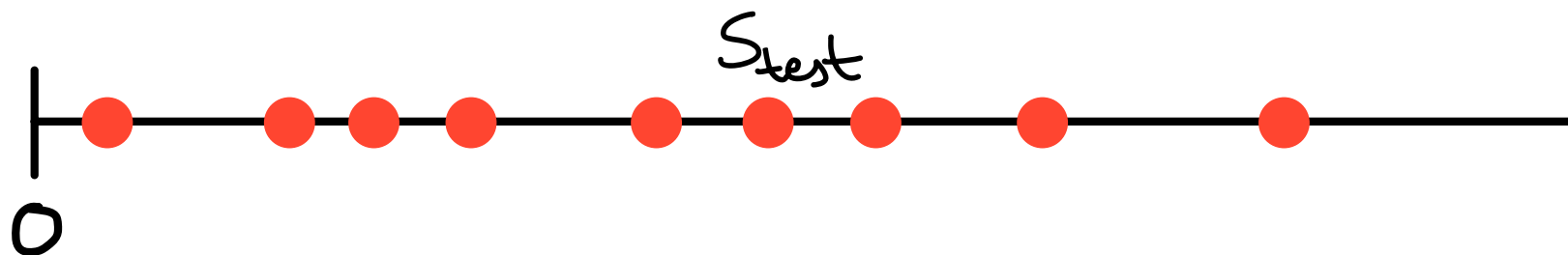
PROOF



⋮



PROOF



$$\mathbb{P}(\text{Rank of } S_{\text{test}} = k)$$

$$= \frac{n!}{(n+1)!} = \frac{1}{n+1}$$

$$\Rightarrow \text{Rank of } S_{\text{test}} \sim \text{Unif}(\{1, \dots, n+1\})$$

PROOF

Rank of $S_{\text{test}} \sim \text{Unif}(\{1, \dots, n+1\})$

$$\Rightarrow \mathbb{P}(\text{Rank of } S_{\text{test}} \leq K) = \frac{K}{n+1}$$

We choose the smallest K s.t.

$$\frac{K}{n+1} \geq 1 - \alpha \quad \text{i.e.} \quad K = \lceil (n+1)(1-\alpha) \rceil$$

Too Good To BE TRUE?



$\hat{f}$ bad

$\Rightarrow \hat{C}_\alpha$ very large

ADVANTAGES

- Post-hoc
- Distribution-free
- Minimal hypotheses
- Finite-sample guarantee

LIMITATIONS

Calibration- marginal guarantee:

$$\mathbb{P} \left(Y_{n+1} \in \underbrace{\hat{C}_\alpha(X_{n+1})}_{\text{how is this set constructed?}} \right) \geq 1 - \alpha$$

how is this set
constructed?

LIMITATIONS

Calibration-marginal guarantee:

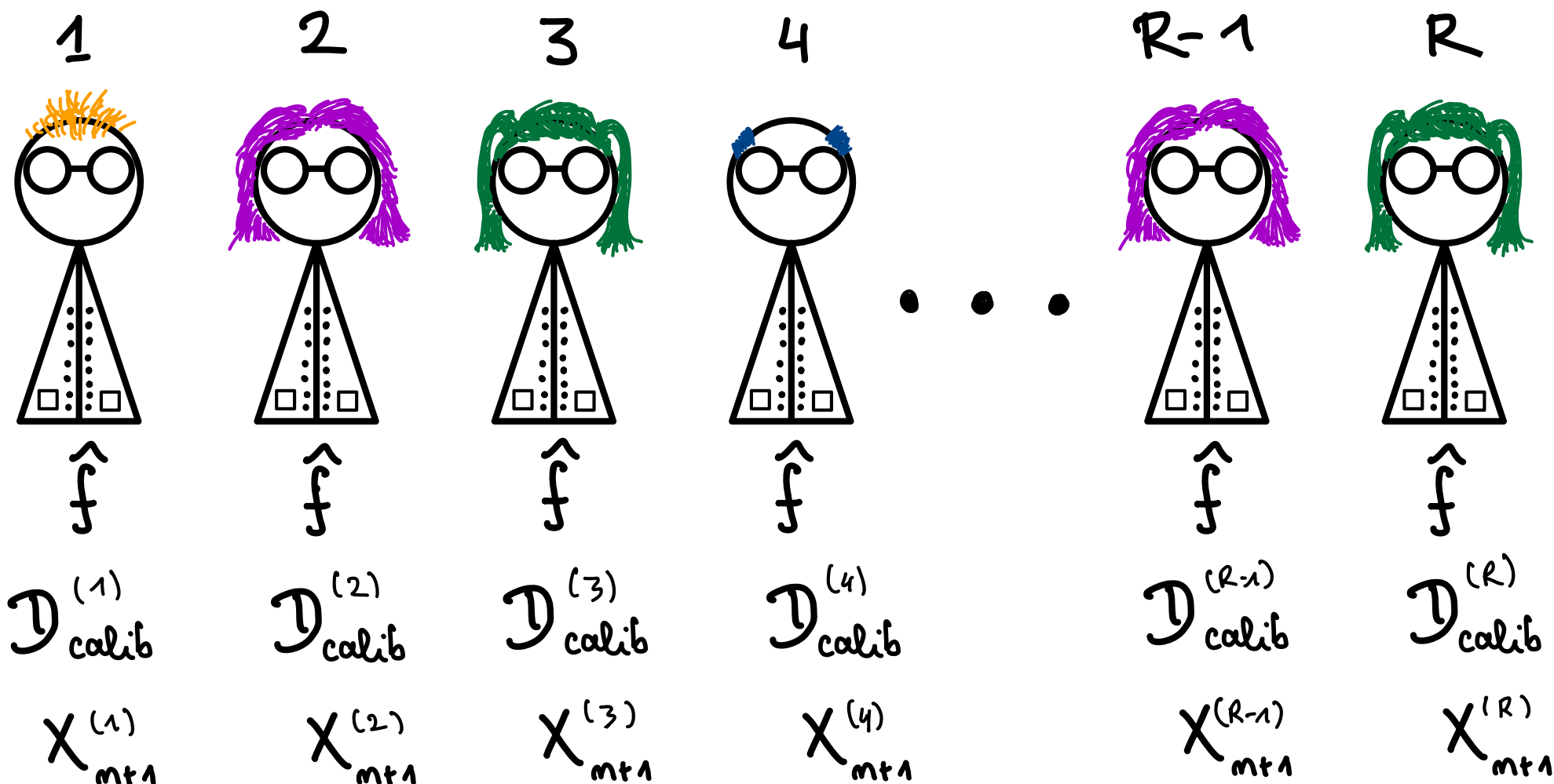
$$\mathbb{P} \left(Y_{n+1} \in \underbrace{\hat{C}_\alpha(X_{n+1})} \right) \geq 1-\alpha$$

how is this set
constructed?

$$\mathbb{P} \left(Y_{n+1} \in \hat{C}(X_{n+1}) \mid X_1, \dots, X_n \right) \geq 1-\alpha ?$$

LIMITATIONS

Calibration-marginal guarantee:

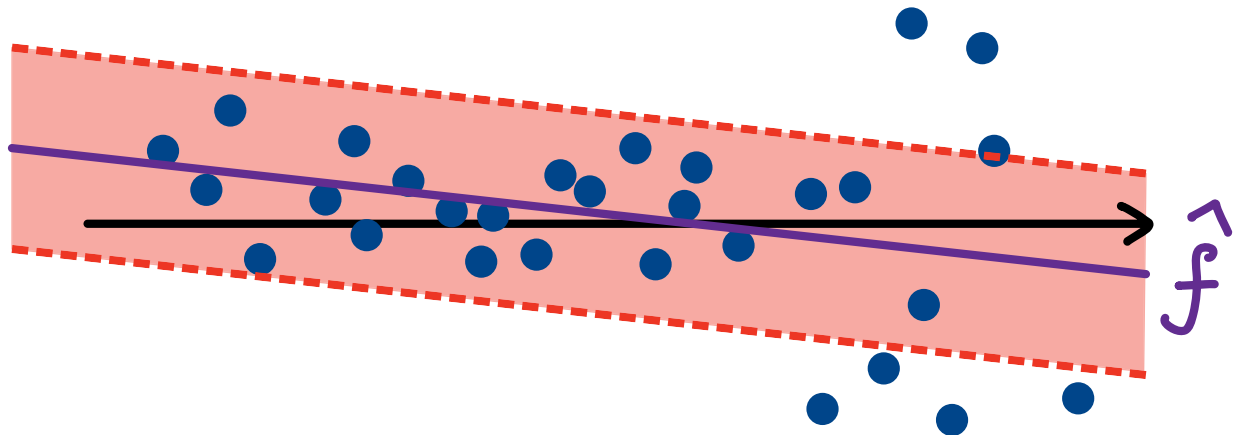


LIMITATIONS

Non-conditional guarantee

$$\mathbb{P}(Y_{n+1} \in \hat{C}_\alpha(X_{n+1})) \geq 1-\alpha \quad \checkmark$$

$$\mathbb{P}(Y_{n+1} \in \hat{C}_\alpha(X_{n+1}) | X_{n+1} = x) \geq 1-\alpha \quad \times$$



SPLIT CONFORMAL

Theorem .- Given a calibration set

$\{(x_i, y_i)\}_{i=1}^n$ and a test point (x_{n+1}, y_{n+1}) ,

if the scores $S_i = |y_i - \hat{f}(x_i)|$ are exchangeable, then the interval

$$\hat{C}_\alpha(x_{n+1}) = [\hat{f}(x_{n+1}) - \delta_\alpha, \hat{f}(x_{n+1}) + \delta_\alpha]$$

satisfies:

$$\mathbb{P}(y_{n+1} \in \hat{C}_\alpha(x_{n+1})) \geq 1 - \alpha$$

SPLIT CONFORMAL

$\hat{f}$ model (obtained by optimizing
on $\mathcal{D}^{\text{train}}$)

$\mathcal{D}^{\text{calib}} = \{ (x_i, y_i) \}_{i=1}^n$ calibration
data

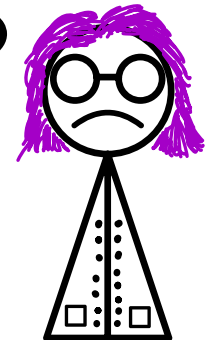
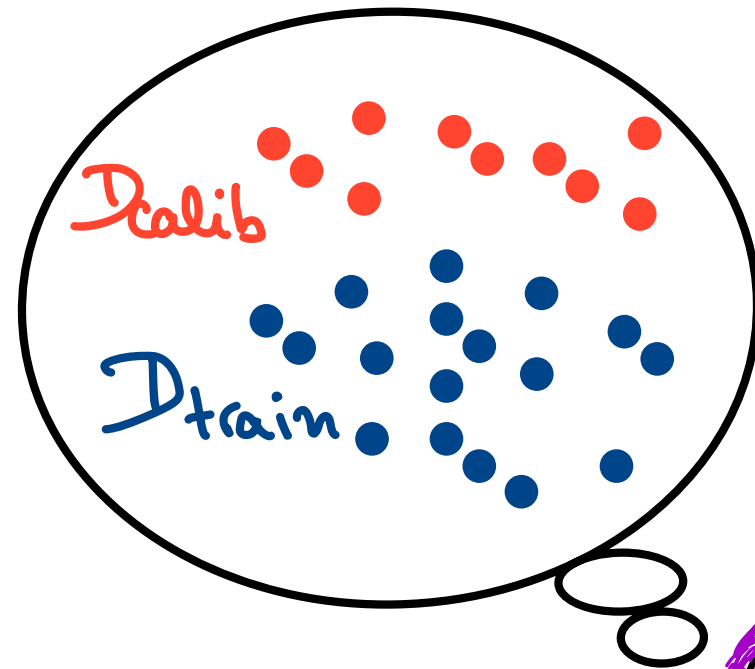
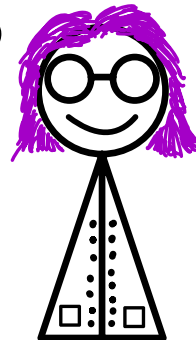
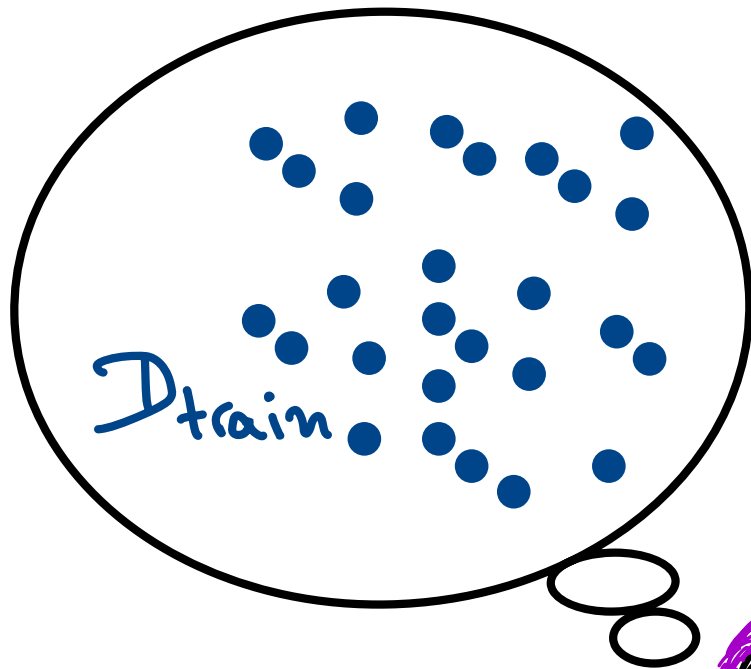
(x_{n+1}, y_{n+1}) test point.



$S_i = |y_i - \hat{f}(x_i)|, i=1, \dots, n+1$
exchangeable

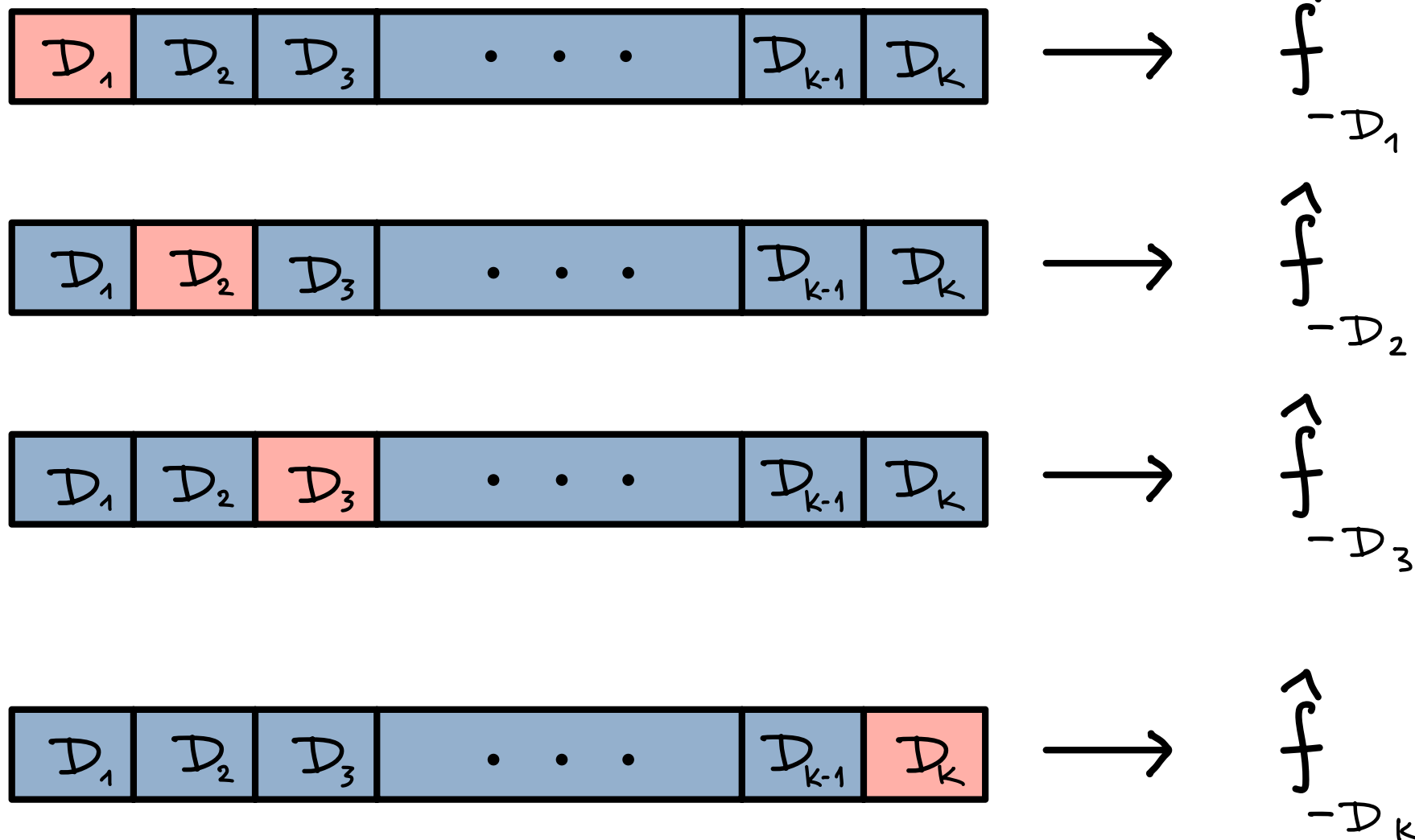
LIMITATIONS

Need for calibration data



JACKKNIFE+ , CROSS-VALIDATION+

Partition the data:



JACKKNIFE+, CROSS-VALIDATION+

Calibration:

$$S_i^{cv} = |Y_i - \hat{f}_{-D_{ind(i)}}(X_i)|, \quad i = 1, \dots, n$$

JACKKNIFE+, CROSS-VALIDATION+

Inference:

$\hat{\ell}_\alpha(x) := \lfloor \alpha(n+1) \rfloor$ -th smallest value of
 $\hat{f}_{-S_{\text{ind}(1)}}(x) - S_1^{\text{cv}}, \dots, \hat{f}_{-S_{\text{ind}(n)}}(x) - S_n^{\text{cv}}$

$\hat{u}_\alpha(x) := \lceil (1-\alpha)(n+1) \rceil$ -th smallest value of
 $\hat{f}_{-S_{\text{ind}(1)}}(x) + S_1^{\text{cv}}, \dots, \hat{f}_{-S_{\text{ind}(n)}}(x) + S_n^{\text{cv}}$

$$\hat{C}_\alpha(x) = [\hat{\ell}_\alpha(x), \hat{u}_\alpha(x)]$$

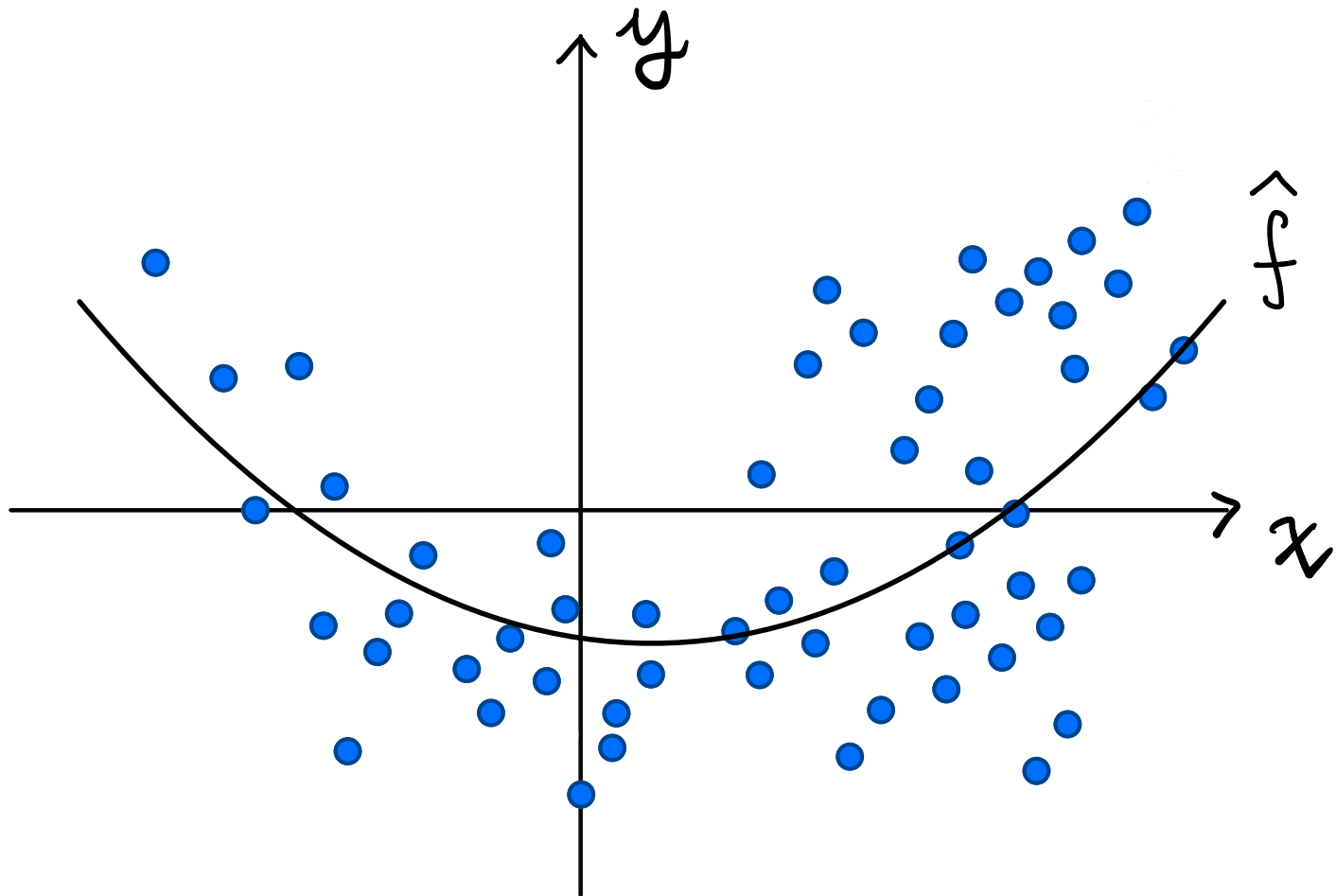
JACKKNIFE+, CROSS-VALIDATION+

Guarantee:

$$\mathbb{P}(Y_{n+1} \in \hat{C}_\alpha(X_{n+1})) \geq 1 - 2\alpha$$

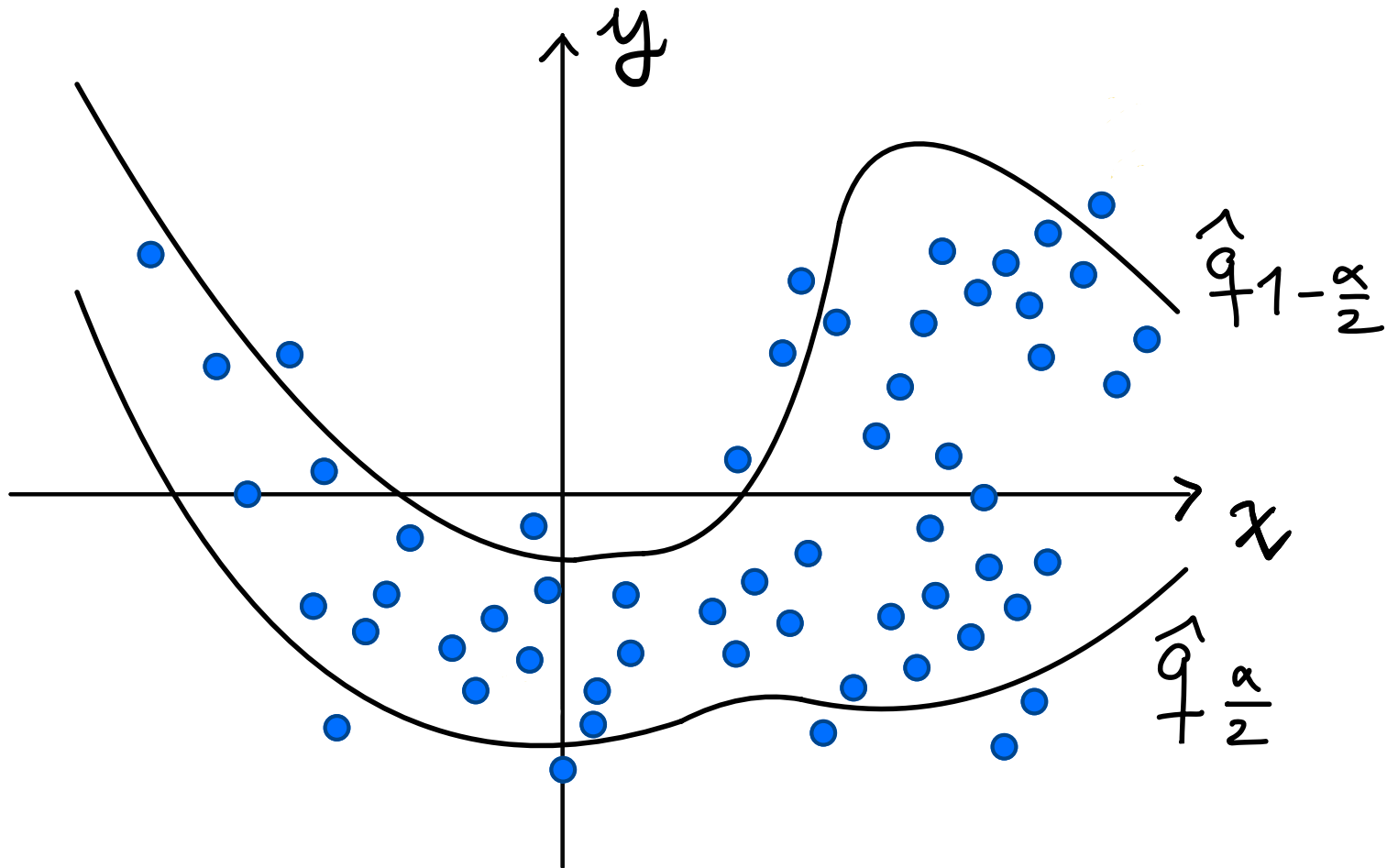
CQR: CONFORMAL

QUANTILE REGRESSION



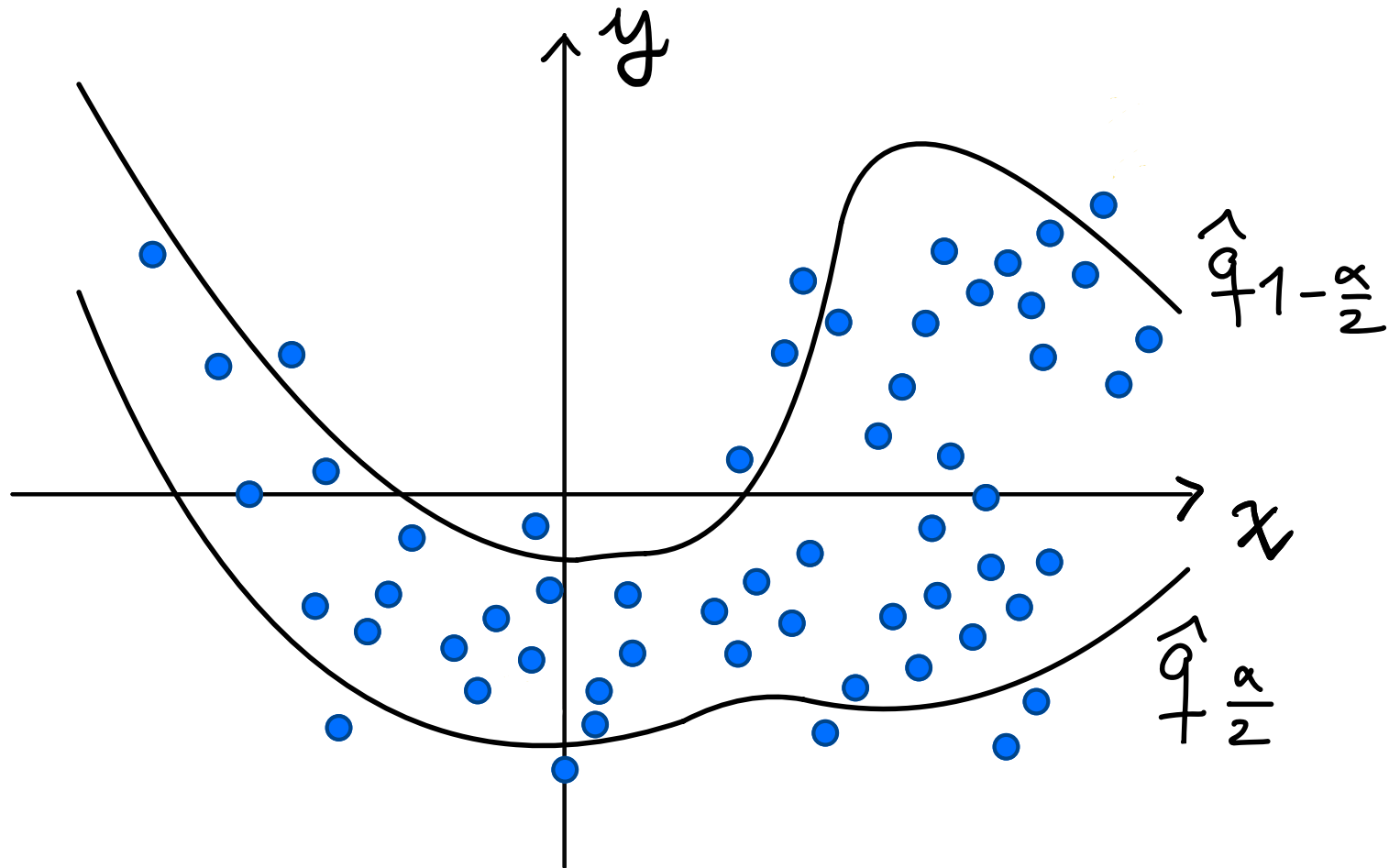
CQR

$$\hat{q}_t(x) \simeq \mathbb{P}(Y \leq t \mid X=x)$$



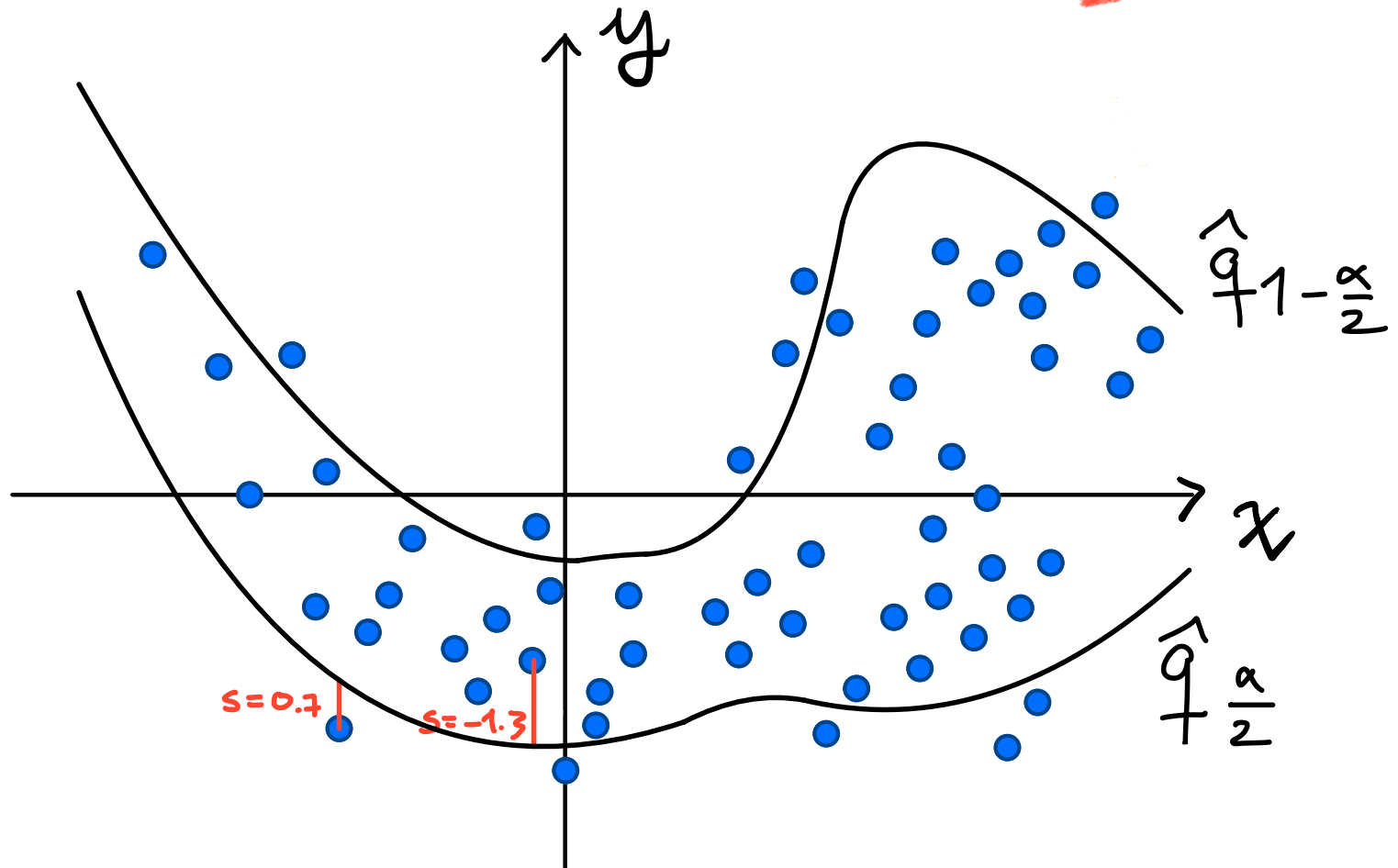
CQR

$$\mathbb{P}(Y \in [q_{\frac{\alpha}{2}}(x), q_{1-\frac{\alpha}{2}}(x)]) = ?$$



CQR: CALIBRATION

$$S_i = \max \left\{ \hat{q}_{\frac{\alpha}{2}}(x_i) - y_i, y_i - \hat{q}_{1-\frac{\alpha}{2}}(x_i) \right\}$$



CQR: CALIBRATION

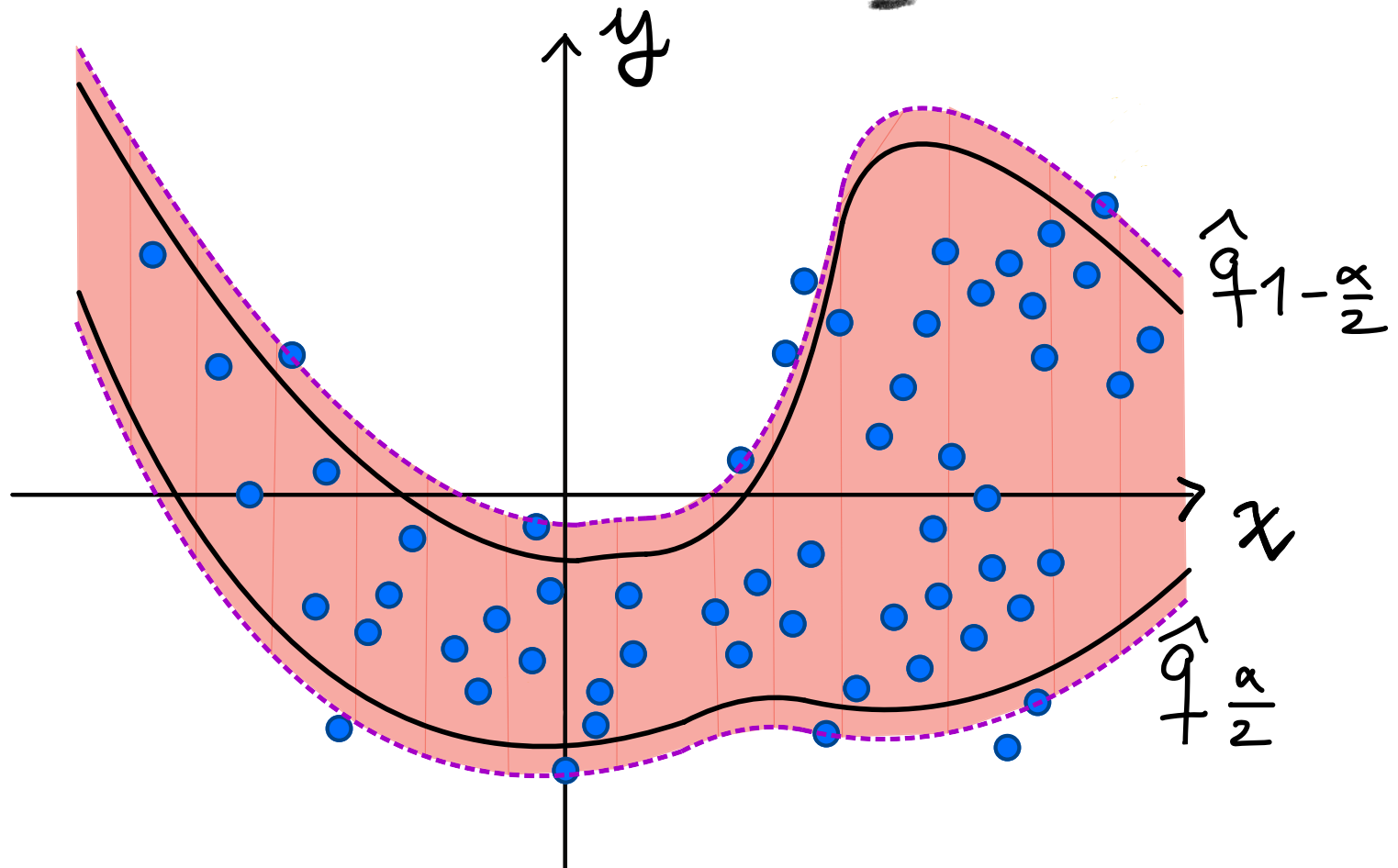
Calibration:

$$S_\alpha := \text{the } \left\lceil \frac{(n+1)(1-\alpha)}{n} \right\rceil - \text{th}$$

quantile of the scores $S_1, \dots, S_n$

CQR: CALIBRATION

$$C_\alpha(x) = [\hat{q}_{\frac{\alpha}{2}}(x) - \delta_\alpha, \hat{q}_{1-\frac{\alpha}{2}}(x) + \delta_\alpha]$$



GENERAL FORMULATION

Given : a predictor $\hat{f}: \mathcal{X} \rightarrow \mathcal{Y}$
a nominal error rate α
a scoring function $s: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$
e.g. in regression $s(y, y') = |y - y'|$

Split conformal: Calibration set $\{(x_i, y_i)\}_{i=1}^n$

$\Rightarrow$ non-conformity scores $R_i = s(\hat{f}_i, y_i)$

$\Rightarrow q_\alpha = \left\lceil \frac{(1-\alpha)(n+1)}{n} \right\rceil$ -quantile of the $\{R_i\}_{i=1}^n$

$\Rightarrow \mathcal{C}(\alpha) := \{y: s(\hat{f}, y) \leq q_\alpha\}$

CONFORMAL CLASSIFICATION

- Least Ambiguous Set-Valued Classifiers (LAC)
- Adaptive Prediction Sets (APS)
- Regularized Adaptive Prediction Sets (RAPS)

LAC

Given : $\hat{\pi}$ softmax classifier
 α nominal error rate

Use the nonconformity scores

$$s(\hat{\pi}, y) = \underbrace{1 - \hat{\pi}_y}$$

i.e. 1 - the probability
of the true label.

APS

Given : $\hat{\pi}$ softmax classifier
 $\propto$ nominal error rate

Rank: $\hat{\pi}_{(1)}(x) \geq \hat{\pi}_{(2)}(x) \geq \dots \geq \hat{\pi}_{(K)}(x)$

$$L(x, \hat{\pi}, z) := \min_{c \in \{1, \dots, K\}} \{ \hat{\pi}_{(1)}(x) + \dots + \hat{\pi}_c(x) \geq z \}$$

APS

Calibration:

$$\hat{\pi}(x_1) \quad \begin{array}{|c|c|c|c|c|} \hline 2 & 5 & 1 & 3 & 4 \\ \hline \end{array} \quad \gamma_1 = 5$$

$\underline{\hspace{1.5cm}} S_1 \underline{\hspace{1.5cm}}$

$$\hat{\pi}(x_2) \quad \begin{array}{|c|c|c|c|c|} \hline 1 & 2 & 4 & 5 & 3 \\ \hline \end{array} \quad \gamma_2 = 1$$

$\underline{\hspace{1.5cm}} S_2 \underline{\hspace{1.5cm}}$

$$\hat{\pi}(x_3) \quad \begin{array}{|c|c|c|c|c|} \hline 5 & 3 & 4 & 1 & 2 \\ \hline \end{array} \quad \gamma_3 = 4$$

$\underline{\hspace{1.5cm}} S_3 \underline{\hspace{1.5cm}}$

$$\hat{\pi}(x_n) \quad \begin{array}{|c|c|c|c|c|} \hline 2 & 3 & 5 & 1 & 4 \\ \hline \end{array} \quad \gamma_n = 2$$

$\underline{\hspace{1.5cm}} S_n \underline{\hspace{1.5cm}}$

APS

Calibration:

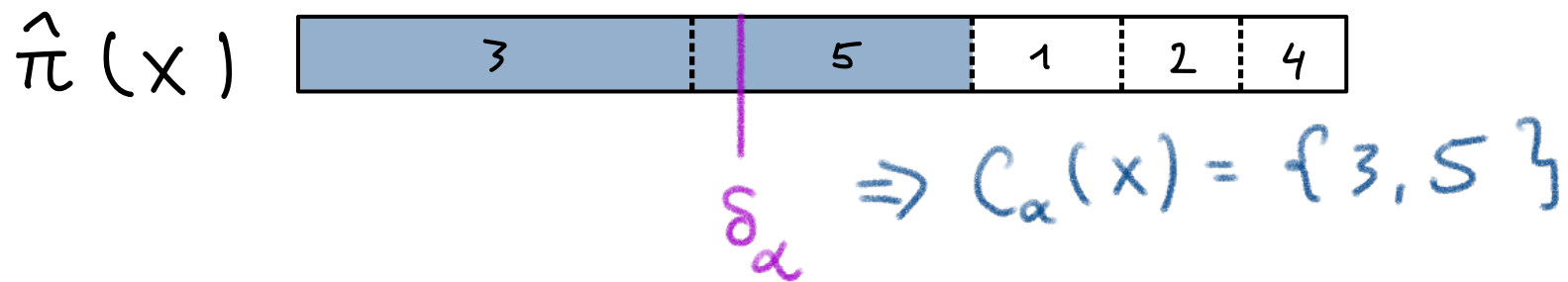
$S_\alpha :=$ the $\left\lceil \frac{(n+1)(1-\alpha)}{n} \right\rceil$ -th

quantile of the scores $S_1, \dots, S_n$

APS

Inference: $\hat{C}_\alpha(x) = \{ (1), (2), \dots, (K) \}$

where $K = \min \{ i : \hat{\pi}_{(1)} + \dots + \hat{\pi}_{(i)} \geq \delta_\alpha \}$



Guarantee:

$$\mathbb{P}(Y_{n+1} \in \hat{C}_\alpha(X_{n+1})) \geq 1 - \alpha$$

APS WITH RANDOMIZATION



δ_α

$u \leftarrow$ Uniform random variable in

$[\pi_3, \pi_3 + \pi_5]$

$$\Rightarrow \hat{\mathcal{C}}_\alpha(x) := \begin{cases} \{3\} & \text{if } u \leq \delta_\alpha \\ \{3, 5\} & \text{if } u > \delta_\alpha \end{cases}$$



Same guarantee, tighter prediction sets!

RAPS

Given : $\hat{\pi}$ softmax classifier
 α nominal error rate

Rank: $\hat{\pi}_{(1)}(x) \geq \hat{\pi}_{(2)}(x) \geq \dots \geq \hat{\pi}_{(x)}(x)$

$$S_i := \hat{\pi}_{(1)} + \dots + \hat{\pi}_{(k)} + \lambda (K - K_{\text{reg}} + 1)$$

K such that

$$(k) = \gamma_i$$

rank of γ_i

hyper-parameters

RAPS

Prediction set:

$$\hat{\mathcal{C}}_a(x) = \{(1), \dots, (K)\}$$

where

$$K = \max \{i : \pi_{(1)} + \dots + \pi_{(i)} + \lambda(i - K_{\text{reg}} + 1) \leq \delta_a\} + 1$$



More stable than APS if softmax is "noisy" i.e. many classes w. low proba.